

PRACTICAL -10

B.DHARSHAN(2520090088)

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dhars@DHARSHAN: /mnt/c/L
Microsoft Windows [Version 10.0.26300.9550]
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C:\Users\dhars>WSL
dhars@DHARSHAN: /mnt/c/Users/dhars$ echo "Operating Systems Practical 10" > original.txt
dhars@DHARSHAN: /mnt/c/Users/dhars$ cat original.txt
Operating Systems Practical 10
dhars@DHARSHAN: /mnt/c/Users/dhars$ ls -i original.txt
15481123719388616 original.txt
dhars@DHARSHAN: /mnt/c/Users/dhars$ stat original.txt
  File: original.txt
  size: 31          Blocks: 0          IO Block: 4096   regular file
Device: 0,69   Inode: 15481123719388616   Links: 1
Access: (0777/-rwxrwxrwx)  Uid: ( 1000/   dhars)   Gid: ( 1000/   dhars)
Access: 2026-10-07 03:38:13.479938400 +0000
Modify: 2026-10-07 03:38:12.373647800 +0000
Change: 2026-10-07 03:38:12.373647800 +0000
 Birth: -
dhars@DHARSHAN: /mnt/c/Users/dhars$ ln original.txt hardlink.txt
dhars@DHARSHAN: /mnt/c/Users/dhars$ ls -li original.txt hardlink.txt
15481123719388616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 hardlink.txt
15481123719388616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 original.txt
dhars@DHARSHAN: /mnt/c/Users/dhars$ ln -s original.txt symlink.txt
dhars@DHARSHAN: /mnt/c/Users/dhars$ ls -li original.txt hardlink.txt symlink.txt
15481123719388616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 hardlink.txt
15481123719388616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 original.txt
4222124650751138 lrwxrwxrwx 1 dhars dhars 12 Oct  7 03:39 symlink.txt -> original.txt
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dhars@DHARSHAN: /mnt/c/Users/dhars$ ls -li original.txt hardlink.txt symlink.txt
15481123719388616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 hardlink.txt
15481123719388616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 original.txt
4222124650751138 lrwxrwxrwx 1 dhars dhars 12 Oct  7 03:39 symlink.txt -> original.txt
dhars@DHARSHAN: /mnt/c/Users/dhars$ stat original.txt
  File: original.txt
  size: 31          Blocks: 0          IO Block: 4096   regular file
Device: 0,69   Inode: 15481123719388616   Links: 2
Access: (0777/-rwxrwxrwx)  Uid: ( 1000/   dhars)   Gid: ( 1000/   dhars)
Access: 2026-10-07 03:38:56.161338000 +0000
Modify: 2026-10-07 03:38:12.373647800 +0000
Change: 2026-10-07 03:38:55.792401100 +0000
 Birth: -
dhars@DHARSHAN: /mnt/c/Users/dhars$ stat hardlink.txt
  File: hardlink.txt
  size: 31          Blocks: 0          IO Block: 4096   regular file
Device: 0,69   Inode: 15481123719388616   Links: 2
Access: (0777/-rwxrwxrwx)  Uid: ( 1000/   dhars)   Gid: ( 1000/   dhars)
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  File: hardlink.txt
  size: 31          Blocks: 0          IO Block: 4096   regular file
Device: 0,69   Inode: 15481123719388616   Links: 2
Access: (0777/-rwxrwxrwx)  Uid: ( 1000/   dhars)   Gid: ( 1000/   dhars)
Access: 2026-10-07 03:38:56.161338000 +0000
Modify: 2026-10-07 03:38:12.373647800 +0000
Change: 2026-10-07 03:38:55.792401100 +0000
 Birth: -
dhars@DHARSHAN: /mnt/c/Users/dhars$ stat symlink.txt
  File: 'symlink.txt' -> 'original.txt'
  size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 0,69   Inode: 4222124650751138   Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/   dhars)   Gid: ( 1000/   dhars)
Access: 2026-10-07 03:39:13.418905800 +0000
Modify: 2026-10-07 03:39:13.418905800 +0000
Change: 2026-10-07 03:39:13.418905800 +0000
 Birth: -
dhars@DHARSHAN: /mnt/c/Users/dhars$ stat -L symlink.txt
  File: symlink.txt
  size: 31          Blocks: 0          IO Block: 4096   regular file
Device: 0,69   Inode: 15481123719388616   Links: 2
Access: (0777/-rwxrwxrwx)  Uid: ( 1000/   dhars)   Gid: ( 1000/   dhars)
Access: 2026-10-07 03:38:56.161338000 +0000
Modify: 2026-10-07 03:38:12.373647800 +0000
Change: 2026-10-07 03:38:55.792401100 +0000
 Birth: -
dhars@DHARSHAN: /mnt/c/Users/dhars$ find . -inum $(ls -i original.txt | awk '{print $1}')
hhh
jj
jj
j
j
j
j
j
j
j
j
j
j
^C
dhars@DHARSHAN: /mnt/c/Users/dhars$ ls -li original.txt
15481123719388616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 original.txt
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dhars@DHARSHAN:/mnt/c/Users/dhars$ ls -li original.txt
15481123719380616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 original.txt
dhars@DHARSHAN:/mnt/c/Users/dhars$ ls -li hardlink.txt
15481123719380616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 hardlink.txt
dhars@DHARSHAN:/mnt/c/Users/dhars$ ls -li symlink.txt
4222124650751138 lrwxrwxrwx 1 dhars dhars 12 Oct  7 03:39 symlink.txt -> original.txt
dhars@DHARSHAN:/mnt/c/Users/dhars$ find . -maxdepth 1 -inum $(stat -c %i original.txt)
./hardlink.txt
./original.txt
dhars@DHARSHAN:/mnt/c/Users/dhars$ ls -li original.txt hardlink.txt symlink.txt
15481123719380616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 hardlink.txt
15481123719380616 -rwxrwxrwx 2 dhars dhars 31 Oct  7 03:38 original.txt
4222124650751138 lrwxrwxrwx 1 dhars dhars 12 Oct  7 03:39 symlink.txt -> original.txt
dhars@DHARSHAN:/mnt/c/Users/dhars$ nano mmap_program.c
dhars@DHARSHAN:/mnt/c/Users/dhars$ gcc mmap_program.c -o mmap_program
dhars@DHARSHAN:/mnt/c/Users/dhars$ ./mmap_program
Original file content: Hello from mmap system call!
Modified file content: hello from mmap system call!
Memory-mapped file operation completed successfully.
dhars@DHARSHAN:/mnt/c/Users/dhars$ cat mmap.txt
hello from mmap system call!dhars@DHARSHAN:/mnt/c/Users/dhars$ nano mmap_prohano mmap_program.c
dhars@DHARSHAN:/mnt/c/Users/dhars$ gcc mmap_program.c -o mmap_program
dhars@DHARSHAN:/mnt/c/Users/dhars$ ./mmap_program
Original file content: Hello from mmap system call!
Modified file content: hello from mmap system call!
Memory-mapped file operation completed successfully.
dhars@DHARSHAN:/mnt/c/Users/dhars$ ~|

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